

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0019

M105 138 kV
March 22, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0019. The Interconnection Customer proposes a 118.4 MW solar photovoltaic facility interconnecting at the M105 138 kV substation in the Pennsylvania area. The request was received on March 10, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 15 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$125,634,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Five Channels-Messick 138 kV
Contingency: Metro Ctr-Dundee 138 kV
Loading: 134.5% (Rating: 189 MVA)
DFAX: 7.4%
Pre-project loading: 129.8%
2. Facility: Hollister-El Sobrante 138 kV
Contingency: Pineville-East Hagerstown 138 kV
Loading: 127.7% (Rating: 238 MVA)
DFAX: 6.2%
Pre-project loading: 124.6%
3. Facility: Cameron-Pluess 138 kV
Contingency: Dundee-Metro Ctr 138 kV
Loading: 127.2% (Rating: 236 MVA)
DFAX: 7.4%
Pre-project loading: 123.5%
4. Facility: Otsego-Carl Bailey 138 kV Line
Contingency: East Hagerstown-Pineville 138 kV
Loading: 125.6% (Rating: 277 MVA)
DFAX: 21.3%
Pre-project loading: 116.5%
5. Facility: M105-Otsego 138 kV
Contingency: W Waynesboro-Chambersburg 345 kV
Loading: 124.8% (Rating: 328 MVA)
DFAX: 47.3%
Pre-project loading: 107.7%

6. Facility: Innerarity-Maryland Solar 138 kV line
Contingency: Pineville-E Hagerstown 138 kV
Loading: 124.6% (Rating: 267 MVA)
DFAX: 6.2%
Pre-project loading: 121.8%
7. Facility: East Agerstown-Pineville 138 kV
Contingency: Innerarity-Pineville 138 kV
Loading: 123.8% (Rating: 346 MVA)
DFAX: 8.0%
Pre-project loading: 121.1%
8. Facility: Maryland Solar-Bellsfork 138 kV
Contingency: East Hagerstown-Pineville 138 kV
Loading: 121.5% (Rating: 274 MVA)
DFAX: 6.2%
Pre-project loading: 118.9%
9. Facility: Bon Aqua-Carl Bailey Substation 138 kV Line
Contingency: Pineville-East Hagerstown 138 kV line
Loading: 121.5% (Rating: 279 MVA)
DFAX: 21.3%
Pre-project loading: 112.5%
10. Facility: Bon Aqua-Dry Gap 138 kV
Contingency: Pineville-East Hagerstown 138 kV
Loading: 120.5% (Rating: 277 MVA)
DFAX: 21.3%
Pre-project loading: 111.3%
11. Facility: Reid Substation-Palgen 138 kV
Contingency: West Waynesboro-Pineville 345 kV Line
Loading: 120.1% (Rating: 335 MVA)
DFAX: 10.1%
Pre-project loading: 116.5%
12. Facility: Antietam-Pineville 138 kV L
Contingency: W Waynesboro-Pineville Substation 345 kV
Loading: 119.4% (Rating: 265 MVA)
DFAX: 10.1%
Pre-project loading: 114.9%
13. Facility: Athalmer Tap-E Hagerstown 138 kV
Contingency: Pineville-Innearity 138 kV
Loading: 118.3% (Rating: 335 MVA)
DFAX: 8.0%
Pre-project loading: 115.4%
14. Facility: Thurmont-Dry Gap 138 kV
Contingency: W Kelso-Dry Gap 138 kV
Loading: 116.7% (Rating: 245 MVA)
DFAX: 16.9%
Pre-project loading: 108.5%
15. Facility: Pluess Substation-Messick 138 kV
Contingency: Dundee-Metro Center 138 kV L
Loading: 116.5% (Rating: 242 MVA)
DFAX: 7.4%
Pre-project loading: 112.8%
16. Facility: East Waynesboro-West Waynesboro 138 kV line
Contingency: Chambersburg-W Waynesboro 345 kV

- Loading: 116.2% (Rating: 266 MVA)
DFAX: 19.0%
Pre-project loading: 107.8%
17. Facility: Cameron-Athlmer Tap 138 kV
Contingency: Dundee-Metro Center 138 kV
Loading: 114.5% (Rating: 273 MVA)
DFAX: 7.4%
Pre-project loading: 111.2%
18. Facility: El Sobrante Substation-Bellsfork 138 kV
Contingency: East Hagerstown-Pineville Substation 138 kV
Loading: 113.6% (Rating: 287 MVA)
DFAX: 6.2%
Pre-project loading: 111.1%
19. Facility: Hollister-Metro Ctr 138 kV
Contingency: E Hagerstown Substation-Pineville 138 kV
Loading: 113.3% (Rating: 256 MVA)
DFAX: 6.2%
Pre-project loading: 110.4%
20. Facility: Reid-Clawson 138 kV
Contingency: W Waynesboro-Pineville 345 kV
Loading: 113.2% (Rating: 341 MVA)
DFAX: 10.1%
Pre-project loading: 109.6%
21. Facility: E Waynesboro-Otsego 138 kV
Contingency: West Waynesboro-Chambersburg 345 kV
Loading: 112.8% (Rating: 304 MVA)
DFAX: 19.0%
Pre-project loading: 105.4%
22. Facility: Antietam-Clawson 138 kV
Contingency: W Wayneboro-Pineville 345 kV line
Loading: 111.5% (Rating: 314 MVA)
DFAX: 10.1%
Pre-project loading: 107.7%
23. Facility: Dundee-Metro Ctr 138 kV L
Contingency: Cameron-Athalmer Tap 138 kV
Loading: 110.0% (Rating: 209 MVA)
DFAX: 5.5%
Pre-project loading: 106.9%
24. Facility: Firstenergy R Paul Smith Power Sta-Dam No. 4 Hydro Sta 138 kV
Contingency: Pineville-Est Hagerstown 138 kV
Loading: 109.8% (Rating: 197 MVA)
DFAX: 7.4%
Pre-project loading: 105.3%
25. Facility: Lamont-Jack McDonough 138 kV L
Contingency: Air Force Base Substation-Aresia Country Club 345 kV Line
Loading: 108.5% (Rating: 203 MVA)
DFAX: 5.2%
Pre-project loading: 105.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Five Channels-Messick 138 kV	Rebuild line with higher-capacity conductor	\$3,935,000	100% (\$3,935,000)
Hollister-El Sobrante 138 kV	Rebuild line with higher-capacity conductor	\$5,596,000	100% (\$5,596,000)
Cameron-Pluess 138 kV	Rebuild line with higher-capacity conductor	\$1,972,000	100% (\$1,972,000)
Otsego-Carl Bailey 138 kV Line	Rebuild line with higher-capacity conductor	\$4,292,000	100% (\$4,292,000)
M105-Otsego 138 kV	Rebuild line with higher-capacity conductor	\$4,414,000	100% (\$4,414,000)
Inerarity-Maryland Solar 138 kV line	Rebuild line with higher-capacity conductor	\$2,842,000	100% (\$2,842,000)
E Hagersown-Pineville 138 kV	Rebuild line with higher-capacity conductor	\$3,321,000	100% (\$3,321,000)
Bellsfork-Mayland Solar 138 kV	Rebuild line with higher-capacity conductor	\$5,411,000	100% (\$5,411,000)
Bon Aqua-Carl Bailey 138 kV	Rebuild line with higher-capacity conductor	\$9,827,000	100% (\$9,827,000)
Bon Aqua-Dry Gap 138 kV	Rebuild line with higher-capacity conductor	\$4,242,000	100% (\$4,242,000)
Reid-Palogen Substation 138 kV line	Rebuild line with higher-capacity conductor	\$903,000	100% (\$903,000)
Antietam-Pineville 138 kV Line	Rebuild line with higher-capacity conductor	\$6,830,000	100% (\$6,830,000)
Athalmer Tap-East Hagerstown 138 kV	Rebuild line with higher-capacity conductor	\$4,863,000	100% (\$4,863,000)
Thurmont-Dry Gap 138 kV	Rebuild line with higher-capacity conductor	\$3,462,000	100% (\$3,462,000)
Messick-Pluess 138 kV	Rebuild line with higher-capacity conductor	\$8,270,000	100% (\$8,270,000)
East Waynesboro-West Waynesboro 138 kV line	Rebuild line with higher-capacity conductor	\$4,929,000	100% (\$4,929,000)
Cameron-Athalmer Tap 138 kV	Rebuild line with higher-capacity conductor	\$4,569,000	100% (\$4,569,000)
El Sobrante-Bellsfork 138 kV	Rebuild line with higher-capacity conductor	\$5,792,000	100% (\$5,792,000)
Hollister-Metro Center 138 kV Line	Rebuild line with higher-capacity conductor	\$3,442,000	100% (\$3,442,000)
Reid-Clawson 138 kV	Rebuild line with higher-capacity conductor	\$6,001,000	100% (\$6,001,000)
E Waynesboro-Otsego 138 kV	Rebuild line with higher-capacity conductor	\$5,362,000	100% (\$5,362,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Clawson-Antietam 138 kV	Rebuild line with higher-capacity conductor	\$2,753,000	100% (\$2,753,000)
Dundee-Metro Center 138 kV	Rebuild line with higher-capacity conductor	\$18,127,000	100% (\$18,127,000)
Firstenergy R Paul Smith Power Sta-Dam No. 4 Hydro Sta 138 kV	Reconductor line	\$1,325,000	100% (\$1,325,000)
Lamont-Jack McDonough 138 kV	Reconductor line	\$3,154,000	100% (\$3,154,000)
	Total allocated to project		\$125,634,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.